UrbanSmartSound: A Methodological Investigation into Urban Sound Classification using Hybrid Architectures and Advanced Augmentation

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Abstract

Modern cities are complex ecosystems, defined not only by their visual topography but also by their soundscape. This paper presents UrbanSmartSound, a systematic investigation into urban sound classification through deep learning. Our goal is to develop a robust classifier for 10 sound classes from the UrbanSound8K dataset, addressing the challenges of background noise and variability in recording devices. Our methodological journey begins with a comparative analysis of three parallel pipelines, based on different feature extraction philosophies: Log-Mel spectrograms, MFCC coefficients, and engineered acoustic features. The results of this initial phase guided the development of an advanced hybrid architecture, which integrates a 2D Convolutional Neural Network (CNN) with a bidirectional Recurrent Neural Network (GRU). We demonstrate how enriching the input with dynamic features (Delta and Delta-Delta MFCCs), optimizing with Focal Loss, and employing the SpecAugment data augmentation technique were decisive for the model's success. Our final classifier achieves an accuracy of 87.0%, proving to be effective and well-balanced. Finally, a critical analysis of One-vs-Rest ensemble architectures reveals the conceptual and performance superiority of the end-to-end multi-class model, which learns in a competitive context essential for discriminating between acoustically similar classes.

1 Introduction

The urban environment is an intricate symphony of sounds. From the constant hum of traffic to the wail of a siren, every acoustic event helps define the soundscape of a city. With the advent of Smart Cities and autonomous systems, the need to equip machines with the ability to "listen" and interpret is emerging. Urban Sound Classification (USC) stands as a key discipline with transformative potential for numerous domains.

An emblematic example is autonomous driving. As highlighted by Walden et al., auditory perception is complementary to visual sensors, as it is not obstructed by the line of sight [7]. A vehicle's ability to instantly recognize the siren of an emergency vehicle, even if not visible, is fundamental for safe navigation. Similarly, intelligent surveillance systems can enhance public safety by automatically detecting anomalous sounds such as gunshots. The challenge, however, is immense: urban sounds are often masked by background noise, distorted, and recorded by microphones of varying quality.

The **UrbanSmartSound** project was born to address these issues through a methodological research path. Our work is not limited to implementing a single architecture but begins with a systematic exploration of three experimental pipelines, each embodying a different signal representation philosophy:

1. **Pipeline A (Log-Mel):** A data-driven approach that treats the Log-Mel spectrogram as an image.
2. **Pipeline B (MFCC):** An evolution that uses a more compact and decorrelated representation of the signal.
3. **Pipeline C (Engineered):** An approach based on the manual engineering of acoustic features.

The comparative analysis of these pipelines allowed us to validate the hypothesis that rich, automatically learned representations were superior. We then focused our efforts on evolving the most promising pipeline, B, transforming it into an advanced hybrid CNN-GRU model. This model was enhanced with dynamic features, a specialized loss function (Focal Loss), and state-of-the-art data augmentation techniques. Finally, we challenged the assumption that a single "generalist" model was the optimal solution by comparing it with a "committee of specialists" (One-vs-Rest ensemble), drawing conclusions about the importance of competitive learning.

2 Methodology

Our methodology is structured to be iterative and comparative, starting from careful data preparation to the exploration of complex architectures.

2.1 UrbanSound8K Dataset

The foundation of our work is the public **UrbanSound8K** dataset [5]. It consists of 8732 labeled audio clips, with a maximum duration of 4 seconds, divided into 10 classes: `air_conditioner`, `car_horn`, `children_playing`, `dog_bark`, `drilling`, `engine_idling`, `gun_shot`, `jackhammer`, `siren`, `street_music`. A key feature of the dataset is its heterogeneity, with recordings at different sample rates and bit depths, reflecting real-world variety. Equally important is its predefined 10-fold split, designed to ensure reproducible experiments and prevent data leakage between training and testing.

2.2 Pre-processing and Data Augmentation

A rigorous pre-processing pipeline was applied to each audio file.

2.2.1 Pre-processing Steps

- **Conversion to Mono:** To reduce computational complexity without a significant loss of discriminant information.
- **Resampling to 22,050 Hz:** A standard compromise between signal quality and computational load, sufficient to represent frequencies up to 11 kHz.
- **Silence Removal (Trimming):** To increase the information density of the signal, ensuring the model focuses on acoustically relevant portions.
- **Amplitude Normalization:** To make the model insensitive to variations in recording volume.
- **Duration Standardization to 4s:** To ensure fixed-size inputs, a fundamental requirement for neural networks.

2.2.2 Data Augmentation Strategies

We employed two complementary augmentation strategies.

Targeted Oversampling (Offline): During the pre-processing phase, we identified the minority classes, `car_horn` and `gun_shot`, and applied targeted oversampling to balance the dataset. For each sample of these classes, new versions were generated by applying random combinations of *pitch shifting*, *time stretching*, and the addition of white noise. This one-time operation ensured that the augmented data remained confined to the fold of the original sample.

SpecAugment (Online): Integrated as the first layer of the model, SpecAugment [4] applies, only during training, random masks directly on the time-frequency input. This technique, as demonstrated by Arnault et al. [1], is extremely effective in improving generalization. The applied masks are:

- **Frequency Masking:** Zeros out a horizontal band of coefficients, forcing the model not to rely on specific frequency bands.
- **Time Masking:** Zeros out a vertical band of timesteps, making the model more robust to temporal occlusions.

2.3 Feature Extraction: From Signal to Image

To make audio data processable by convolution-based architectures, which are designed to operate on grid-like data such as images, it is necessary to transform the one-dimensional audio signal (amplitude vs. time) into a two-dimensional time-frequency representation. This process is fundamental, as the choice of representation directly influences the model's ability to extract discriminant features.

2.3.1 Log-Mel Spectrograms

The first step to obtain a time-frequency representation is to calculate a spectrogram using the Short-Time Fourier Transform (STFT), which breaks down the signal into overlapping time windows and computes the frequency spectrum for each. However, a simple spectrogram with a linear frequency scale does not reflect human auditory perception: the human ear is much more sensitive to variations in low frequencies than in high frequencies. For this reason, we adopted the Mel scale, a perceptual scale that maps linear frequencies (f) to a scale that approximates the response of the human ear, as described by the formula:

$$m = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right) \tag{1}$$

This representation was the input for our Pipeline A.

2.3.2 Mel-Frequency Cepstral Coefficients (MFCCs)

Although the Log-Mel spectrogram is a powerful representation, the energies in adjacent Mel filter banks are often highly correlated. MFCCs, the input for Pipeline B, push this philosophy a step further, aiming to produce an even more compact and decorrelated representation. After obtaining the Log-Mel spectrogram, the Discrete Cosine Transform (DCT) is applied to decorrelate the coefficients. For our advanced model, we stacked the MFCCs with their temporal derivatives, **Delta** (Δ) and **Delta-Delta** ($\Delta\Delta$), creating a 3-channel input that explicitly captures the dynamics of the sound. This tensor not only describes the shape of the spectrum at each moment but also explicitly captures its dynamic evolution, crucial information for distinguishing similar sounds with different temporal patterns (e.g., a stationary sound from a pulsating one).

2.3.3 Engineered Feature Vector (Pipeline C)

In contrast to the data-driven approaches of Pipelines A and B, Pipeline C embodies a more traditional philosophy based on manual feature engineering. The goal was to create a one-dimensional vector that captured various acoustic properties of the signal, which would then be fed into a 1D CNN, suitable for processing multivariate time series. The features extracted for each audio clip are:

- **Chromagram:** Measures the distribution of spectral energy across the 12 chromatic pitch classes. It is particularly useful for capturing the harmonic and melodic characteristics of the sound.
- **Spectral Contrast:** Calculates the amplitude difference between the peaks and valleys of the spectrum. This feature helps distinguish sounds with a clear tonal structure from noisier or more textured sounds.
- **Spectral Centroid:** Represents the "center of mass" of the spectrum, indicating where most of the energy is concentrated. It is an indicator of a sound's "brightness."
- **Spectral Rolloff:** Is the frequency below which a specified percentage (e.g., 85%) of the total spectral energy is found. It provides another measure of the spectrum's shape.
- **Zero Crossing Rate (ZCR):** Indicates the rate at which the audio signal crosses the zero axis. High values are often associated with noisy or percussive sounds.

Assembling the Input Vector. The main challenge was to combine these features, which have different dimensionalities. Chromagram and Spectral Contrast are 2D matrices (with shape *feature* $\times$ *timesteps*), while the others are 1D vectors (with shape $1\times$ *timesteps*). To create a coherent input for the 1D CNN, we performed the following steps:

1. The 1D features were expanded to have a "feature" dimension of 1.
2. All features (the original 2D matrices and the expanded 1D vectors) were concatenated along the feature axis.
3. The resulting matrix was transposed to obtain a final format of **(*timesteps*, *features*)**.

This process produced a multivariate time series, where for each timestep, there is a vector describing the different acoustic properties of the signal at that instant, preserving the temporal sequence for analysis by the 1D CNN.

2.4 Experimental Architectures and Advanced Model

Our journey began with three parallel pipelines and converged on a hybrid architecture.

Hybrid CNN-RNN Architectures. The inspiration to combine CNNs and RNNs comes from established works in the field, such as those by Spoorthy et al. [6] and Wang's thesis [2], which highlight how CNNs excel at extracting local time-frequency features, while RNNs model long-term temporal dependencies.

2.4.1 Recurrent Block: Gated Recurrent Unit (GRU)

Downstream from the CNN, we inserted two layers of bidirectional GRUs [2].
A GRU controls the flow of information through an **update gate** and a **reset gate**.
These mechanisms allow the network to balance between preserving past context and incorporating new input.
This makes GRUs efficient for sequence modeling while reducing computational complexity.
The use of bidirectional GRUs allows the model to process the sequence in both directions, capturing a richer context.

2.4.2 Loss Function: Focal Loss

To address class imbalance and focus training on "hard" examples, we used Focal Loss [3]:

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{2}$$

where the **focusing parameter** γ reduces the weight of easy samples, while α_t balances the importance of the classes. We used $\gamma = 1.5$ and $\alpha = 0.25$.

3 Results and Discussion

Our training process followed a standardized structure for each pipeline, based on the dataset's fold split. The pre-computed data was loaded, and the model was trained and evaluated.

3.1 Phase 1: Comparison of Initial Pipelines

A first training cycle (10 epochs), also following a comparison using a cross-validation technique for each pipeline, revealed a clear performance hierarchy (Table 1).

Table 1: Preliminary performance of the initial pipelines.

Pipeline	Input Features	Accuracy
A (2D-CNN)	Log-Mel Spectrogram	≈ 61%
B (2D-CNN)	MFCC	≈ 65%
C (1D-CNN)	Engineered Vector	≈ 40%

The gap of over 20 percentage points between the data-driven approaches (A, B) and the one based on engineered features (C) confirmed our hypothesis: allowing the network to autonomously learn features from a rich representation is a winning strategy.

3.2 Phase 2: Iterative Optimization and Final Model

The results of Phase 1 unequivocally indicated the superiority of the data-driven approaches, with Pipeline B (based on MFCCs) emerging as the most promising. The second phase of our work was therefore dedicated to a methodological evolution of this pipeline, transforming it from a basic model into an advanced architecture through a process of analysis, hypothesis, and validation. In this phase, we also implemented offline data augmentation across all pipelines, using targeted oversampling to address class imbalance. Specifically, the minority classes *car_ horn* and *gun_ shot* were augmented through random combinations of pitch shifting, time stretching, and white noise addition, ensuring that the new samples remained confined to the folds of the original data.

3.2.1 Impact of the Optimized Architecture on Original Pipelines

Before proceeding with the specific evolution of Pipeline B, a crucial step was to validate the effectiveness of the new base architecture (hybrid 1D/2D-CNN + 2 Bi-GRU, trained with Focal Loss) by retraining the three original pipelines, also increasing the epochs from 10 to 100. The goal was to understand whether the improvements were solely related to the features or if the architecture itself played a decisive role. The results, summarized in Table 2, were illuminating.

Table 2: Performance of the retrained pipelines.

Pipeline	F1-Score (Weighted)	Accuracy
A (Log-Mel)	0.84	84.04%
B (MFCC)	0.83	83.31%
C (Vector)	0.75	74.75%

Table 3: Loss values of the pipelines.

Pipeline	Loss
A (Log-Mel)	0.0996
B (MFCC)	0.0917
C (Vector)	0.0875

This experiment revealed that pipelines A and B showed a remarkable performance jump, moving from a preliminary accuracy of 61-65% to over 83-84%. This demonstrates that much of the improvement was attributable to the robustness of the hybrid architecture and the Focal Loss function, not just the choice of features.
These results solidified our decision to focus the final optimization (the introduction of Deltas and SpecAugment) on the pipeline that offered the best combination of feature richness and baseline performance: Pipeline B.

3.2.2 Development of Pipeline B-Advanced: A Three-Pillar Approach

The goal was to overcome the limitations of a basic model and address systematic confusions between acoustically similar classes. We structured the development of **Pipeline B-Advanced** on three fundamental pillars:

1. **Enrichment of Input Features:** A qualitative analysis of early errors revealed that the model struggled to distinguish sounds with similar spectral textures but different temporal dynamics (e.g., the stationary hum of an `air_conditioner` versus the more variable hum of an `engine_idling`). To provide the model with this crucial information, we replaced the single-channel MFCC input with a 3-channel tensor, stacking the static MFCCs with their temporal derivatives (Delta and Delta-Delta).

2. **Enhancement of the Recurrent Architecture:** Inspired by the literature [2], we hypothesized that a deeper temporal learning hierarchy could better capture complex sequences. We therefore replaced the single Bi-GRU layer with a stacked layer structure, allowing the model to learn temporal features at different levels of abstraction. To mitigate the consequent risk of overfitting, we inserted Dropout layers between the GRU blocks.

3. **Optimization of the Loss Function:** We adopted Focal Loss to force the model to focus on the most difficult samples and resolve systematic confusions between acoustically similar classes, giving less weight to errors on "easy" and already learned examples.

3.2.3 Analysis of the 3-Layer GRU Model: A Success with Side Effects

The first implementation of Pipeline B-Advanced, equipped with **3 Bi-GRU layers**, produced a test set accuracy of **84.87%**. Although this value was marginally higher than that obtained by a base model trained for more epochs (83%), a qualitative analysis of the confusion matrix revealed a fundamental and ambivalent change:

- **Targeted Success:** The critical confusion between the stationary sounds `engine_idling` and `air_conditioner` was almost completely resolved. This validated our hypothesis: the dynamic features and deep recurrent architecture were effective in discriminating sounds based on their temporal evolution.
- **New Criticality (Rhythmic Overfitting):** A new and significant confusion had emerged between the percussive sounds `drilling` and `jackhammer`. The model's precision for the `jackhammer` class had plummeted to an unacceptable **65%**. Our hypothesis was that the 3-GRU model, by becoming extremely sensitive to temporal patterns, had **over-specialized** its learning on marked rhythms, excessively generalizing the jackhammer pattern and erroneously applying it to drilling sounds as well.

3.2.4 The "Sweet Spot": Optimization with 2 GRU Layers

The problem that emerged was not one of pure performance, but of balance and robustness. To solve it, we conducted a targeted optimization experiment, reducing the model's complexity to counteract rhythmic over-specialization. The recurrent component was simplified to **2 Bi-GRU layers**.
The resulting model achieved an accuracy of **84.93%**, virtually identical to the previous one, but with a significantly more balanced classification behavior, as revealed by the analysis:

- The confusion between `drilling` and `jackhammer` was significantly reduced, with precision for `jackhammer` rising to a more robust **73%**.
- The confusion between `engine_idling` and `air_conditioner` slightly increased again, highlighting a clear architectural **trade-off** between the ability to distinguish stationary sounds and the ability to distinguish percussive sounds.

This 2-GRU model represented the best compromise achievable through architectural optimization alone: a "sweet spot" that balanced complexity and generalization capacity.

3.2.5 The Decisive Impact of SpecAugment: Overcoming the Final Plateau

Having identified the optimal base architecture, we addressed the remaining critical issues with a state-of-the-art data augmentation technique, **SpecAugment**, integrated directly into the model. The goal was to force the model to learn more holistic representations that were less dependent on specific local features, both spectral and temporal, thus resolving the observed trade-off.
The introduction of SpecAugment produced the definitive quality leap, bringing the final test set accuracy to **87.0%** (Loss **0.0743**). This result not only exceeded the set target but, as highlighted by the detailed classification report (Table 4), demonstrated the resolution of the previous critical issues. A comparative analysis of the F1-scores highlights the success of the strategy:

- The F1-score for the `drilling` class, the weak point of the previous model, rose to a solid **0.78**.
- Performance on all other critical classes, such as `jackhammer` and the `engine_idling`/`air_conditioner` pair, also improved, leading to an overall more reliable and balanced model.

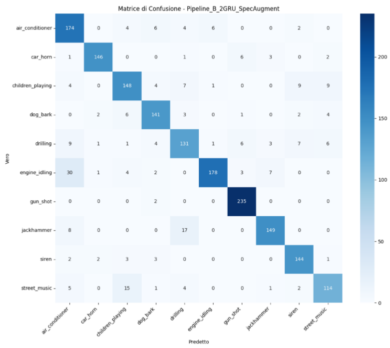

Figure 1: Normalized confusion matrix of the final model on the test set.

The confusion matrix (Figure 1) visually confirms this success, showing a stronger diagonal and less error dispersion among classes. The project had thus demonstrated that the combination of a rich signal representation, a carefully balanced hybrid architecture, and advanced augmentation techniques constitutes a winning strategy for this complex task.

Table 4: Classification report of the final optimized model.

Class	Precision	Recall	F1-Score	Support
air_conditioner	0.75	0.89	0.81	196
car_horn	0.96	0.92	0.94	159
children_playing	0.82	0.81	0.82	182
dog_bark	0.87	0.89	0.88	159
drilling	0.78	0.78	0.78	169
engine_idling	0.96	0.79	0.87	225
gun_shot	0.94	0.99	0.96	237
jackhammer	0.91	0.86	0.88	174
siren	0.87	0.93	0.90	155
street_music	0.84	0.80	0.82	142
Accuracy		**0.870**		**1798**
Weighted Avg	**0.87**	**0.87**	**0.87**	**1798**

3.3 Phase 3: The Ensemble Test – Specialization vs. Competition

After optimizing our multi-class model to an accuracy of 87.0%, the final phase of the project was dedicated to exploring a fundamental question: can a "committee of specialists" (an ensemble architecture) outperform a single "generalist" classifier? The chosen approach was **One-vs-Rest (OvR)**, training 10 binary classifiers, each dedicated to recognizing a single class against all others.

3.3.1 Experiment 1: Simple One-vs-Rest (OvR) Ensemble

The first experiment involved training 10 binary models, based on our optimized architecture (2D-CNN + 2 Bi-GRU + SpecAugment), but with a sigmoid output and binary focal loss. The final ensemble decision was made via a *winner-takes-all* mechanism, selecting the class whose classifier provided the highest probability (argmax).

The results, however, were counterintuitive. The OvR ensemble achieved an overall accuracy of **81.8%**, a value significantly lower than that of the multi-class model. The analysis revealed that, although each binary classifier achieved high accuracy in its isolated task (often >94%), this metric was deceptive. The high performance was mainly due to the ability to correctly classify the vast majority of "easy" negative samples, but it did not reflect a real ability to discriminate between acoustically similar classes. The system suffered from a **loss of competitive context**: each model learned to recognize its own class in isolation, without ever learning to explicitly distinguish it from its closest "rivals."

3.3.2 Experiment 2: Hybrid Heterogeneous Cascading Ensemble

Based on these findings, we designed a second, more sophisticated experiment to mitigate the weaknesses of the OvR approach. The hypothesis was to create a **heterogeneous ensemble** combined with a cascading logic for cases of uncertainty.

- **Architecture Specialization:** Based on the analyses of previous models, for stationary and temporally complex classes (`engine_idling`, `air_conditioner`), binary classifiers with a deeper 3-layer Bi-GRU recurrent network were used. For all other classes, the more balanced 2-layer Bi-GRU architecture was maintained.
- **Cascading Logic:** For samples where the OvR ensemble showed uncertainty (defined as cases where more than one binary classifier returned a probability above an 80% threshold), the final decision was delegated to our best multi-class model, which acted as an "expert" ambiguity resolver.

This advanced hybrid system achieved a final accuracy of **83.15%**. Although this represents a measurable improvement over the basic OvR ensemble, it failed to match the performance of the single multi-class model. The analysis revealed that the cascading logic was triggered for only a small number of samples (1.72% of the test set), indicating that the binary classifiers, while often wrong, tend to produce probability scores that rarely cross the uncertainty threshold competitively.

The detailed classification report (Table 5) and the confusion matrix (Figure 3) confirm the intrinsic limitations of the approach. Despite using a 3-GRU model, the confusion between `engine_idling` and `air_conditioner` (32 errors) remained significant, and the precision for the `jackhammer` class dropped to 68%.

Table 5: Classification report of the Hybrid Ensemble.

Class	Precision	Recall	F1-Score	Support
air_conditioner	0.73	0.79	0.76	196
car_horn	0.96	0.90	0.93	159
children_playing	0.78	0.81	0.79	182
dog_bark	0.89	0.89	0.89	159
drilling	0.71	0.79	0.75	169
engine_idling	0.95	0.73	0.83	225
gun_shot	0.97	0.98	0.98	237
jackhammer	0.68	0.89	0.77	174
siren	0.86	0.85	0.86	155
street_music	0.83	0.64	0.73	142
Accuracy		**0.8315**		**1798**
Weighted Avg	**0.84**	**0.83**	**0.83**	**1798**

3.3.3 Conclusion on Modeling Strategy

The research path, culminating in experiments with ensemble architectures, converges on a fundamental conclusion: for the task of urban sound classification, an end-to-end multi-class model is inherently superior to a set of binary specialists. The ability of a model to learn in a competitive context, where the Softmax function forces a relative decision among all classes, is more effective than isolated specialization. The single model learns to resolve ambiguities, while the ensemble tends to perpetuate them.

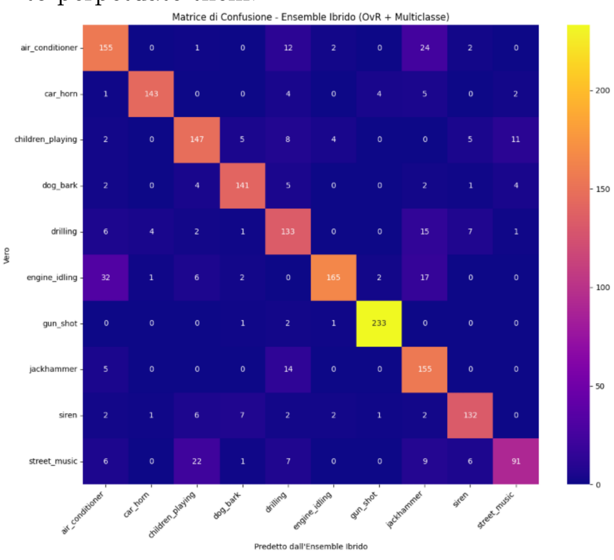

Figure 3: Confusion matrix of the hybrid ensemble. The overall accuracy is 83.15%.

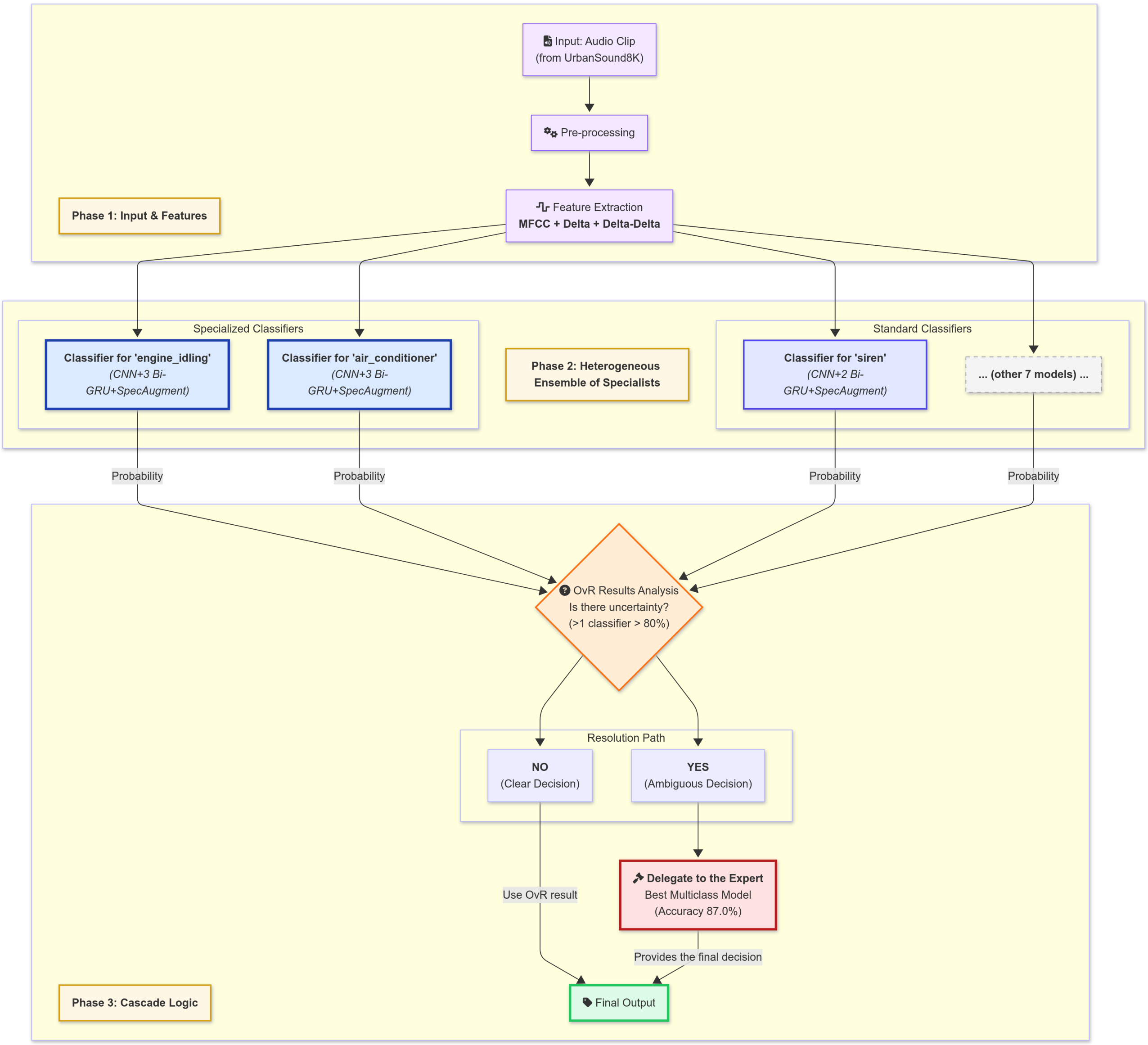

Figure 2: OvR diagram

4 Conclusions

This project has conducted a methodological exploration of urban sound classification, leading to conclusions that are both technical and conceptual.

1. **Representation is crucial:** Data-driven approaches using rich signal representations (MFCCs, Log-Mel) are inherently superior to manual feature engineering for this task.

2. **Dynamics distinguish:** The addition of dynamic features (Delta and Delta-Delta) was fundamental to resolving ambiguities between sounds with similar spectral textures but different temporal evolutions.

3. **Augmentation robustifies:** The integration of SpecAugment proved to be the decisive move to overcome a performance plateau, significantly improving the model's generalization.

4. **Competition wins over specialization:** Our most significant discovery is that, for a task with high acoustic overlap between classes, an end-to-end multi-class model is superior to a set of binary specialists. The need for competitive learning, imposed by the Softmax function, is more effective than isolated specialization.

Ultimately, our study identifies a single hybrid model (2D-CNN + 2 Bi-GRU), fed by MFCCs enriched with their derivatives and regularized by SpecAugment, as the optimal solution. The achieved accuracy of 87.0% is not only an excellent technical result but also the validation of a methodological approach that prioritizes iterative analysis, targeted optimization, and a deep understanding of the model's learning mechanisms.

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